

Mouse auditory experiment

A first plan

October 8, 2026

We want to play short tones to a mouse and record its brain responses. SONIC tested awake sheep; our mouse state and tone settings remain open. (Methods 6.6, p.13)

1. Choose the sounds

- Choose the frequency, duration, loudness, and gap between tones.
- **SONIC:** initially explored 1.4–32 kHz; benchmark tones were 10 ms sine waves with 2 ms fades at each end. (Methods 6.1–6.2, p.11)
- **SONIC:** random frequencies, balanced within each block, with no silence between tones. (Methods 6.1, p.11)
- Save the sequence before playback, as SONIC did. (Methods 6.2, p.11)

2. Set up the speaker

- Plan as though we need to assemble the whole sound setup.
- **SONIC:** NI USB-6361 output at 192 kHz, TDT SA1 amplifier, and MF1 free-field speaker; about 80 dB SPL at sheep-ear distance. (Methods 6.2, p.11)
- **Our plan:** compare the equipment on the next page, keep speaker placement fixed, and measure each tone with a calibrated microphone at the mouse-ear position.

3. Keep everything on one timeline

- Schedule each tone with a timing pulse and record it alongside the neural signals.
- **SONIC:** recorded electrical sound output and trial-start markers; shared clock references aligned these with Connexus brain recordings. (Methods 6.2, p.11)
- **Our additions:** synchronize a microphone recording to check actual sound arrival. Align video frame times too if we add behavior video.

Reference: Perkins et al. (2025), SONIC: A Benchmarking Paradigm for Brain-Computer Interfaces, bioRxiv v1. Page numbers above refer to the printed manuscript. Mouse choices and additions are proposals.

Equipment we need

Mouse setup: building from scratch

Xiaorong forwarded Tinghan's equipment list. We will use it to compare options; access to those items still needs confirming.

1. Play the tones

- **Candidate path:** computer $\rightarrow$ timed sound output (DAC/DAQ) $\rightarrow$ ED1 driver $\rightarrow$ ES1 speaker.
- **ES1:** plays sound through the air. Its intended range starts at 4 kHz; SONIC's lowest tones fall below that range. (TDT speaker specs)
- **ED1:** also needs zBus power, such as ZB1PS, and the matching speaker cable. It does not generate the tones. (TDT driver specs)
- **Other choices:** EC1 delivers sound through tubing. Peerless XT25TG30-04 and Scan-Speak R2004/602200 need a conventional amplifier; check their usable frequency range before choosing.

2. Measure the sound

- Choose **one complete setup:** A (116Hm kit), B (filtered microphone + NI), or C (B&K chain). The measurement page shows what to get together.
- Parts already included in a kit do not need buying again. The other setups are alternatives.
- Each setup needs calibration records and a sound-level reference. Check each tone at the planned ear position.

3. Record the timing and brain signals

- Choose the neural recorder and headstage, then match its event inputs to the tone-output system.
- Record tone markers in both sound and neural data. Check actual sound arrival and clock drift; a shared marker alone does not make two clocks identical.
- Include cables, power, fixed mounts, and data storage. Add a camera only if behavior is needed; align its frame times too.
- A soundproof box is not required by current team direction. Record background sound and the setup geometry.

Equipment candidates are from the forwarded BCM list; linked specifications were checked October 8, 2026. Mouse state, frequency range, and final hardware remain open.

Choose one sound setup

Parts in each group work together; groups are alternatives

Choose **S1, S2, S3a, or S3b**. Get one of each part named in your chosen group.

Choice	Get these together	Parts subtotal
S1: TDT through-air speaker	ES1 + ED1 + ZB1PS power	All need quotes
S1 tubing alternative	EC1 replaces ES1; keep ED1 and ZB1PS	All need quotes
S2: SONIC reference	MF1 + SA1 + ZB1PS power	All need quotes
S3a: conventional tweeter	Peerless XT25TG30-04 (\$28.47) + ART SLA-1 (\$279.99)	\$308.46
S3b: different tweeter	Scan-Speak R2004/602200 (\$178.60) + ART SLA-1 (\$279.99)	\$458.59

Add one timed sound-output device

- **NI USB-6361: \$3,030.42.** SKU 781442-01. Add one waveform-output device for the chosen sound group. (dealer listing)
- If measurement choice B uses this same NI unit, count it **once**, not again as another recording device.

What is separate or optional?

- Check the chosen package's cables and mounts; add only what is missing.
- **AD3: \$379, optional bench tool.** It is not an automatic extra or a verified replacement for the final NI setup. (Digilent)
- These groups remain proposals. ES1's intended range starts at 4 kHz; each chosen speaker needs acoustic and timing checks.

Prices checked October 8, 2026, in USD; tax/shipping extra unless included by the seller. Peerless showed zero stock; Scan-Speak stock is unconfirmed. TDT uses its quote route.

Choose one measurement setup

Choose A, B, or C; do not buy every component listed

A. One complete Avisoft kit

- **Get 1 complete 116Hm kit: \$8,160**, or \$7,320 if eligible for educational pricing. Choose SKU 51165 or 51166.
- **Already included:** microphone, preamplifier, acoustic recorder, cable, stand, and recording software.
- **Do not add:** a separate 116Hm base, CM16 microphone, mic power module, or B&K parts.

B. One microphone package with NI acquisition

- **Get 1 filtered 40026 package: \$2,580, plus 1 compatible NI USB-6361: \$3,030.42.** NI + microphone subtotal: \$5,610.42.
- **Already in 40026:** microphone, preamplifier, filter, cables, and stand. NI records its output; power and acquisition settings need checking.
- Use the **same NI unit** counted for sound output if that configuration is verified. Do not also get the 116Hm kit or the other microphone variants.

C. A specialist B&K setup

- **Get 1 set of 4939-A-011 + 1 matched conditioner + 1 suitable acoustic recorder.** Models and complete-chain quotes still need defining.
- The set includes the 4939 microphone and 2670 preamplifier. Do not add those two parts separately. This is an alternative to A/B.

Calibration is a separate choice

- Add a reference suited to our tone range. The **\$552 generator checks 40 kHz only**; it is an optional reference, not part of every setup.
- A/B's microphone specification starts at 2 kHz. We need another or separately characterized choice if lower tones are required.

A is the most bundled option; B needs more integration. Both still need a computer and timing checks. Kit contents, microphone variants, and USD prices. Detailed component prices are retained in the repository catalog.

Prices: recording and data

Examples for functions we still need to provide

If we choose the Intan example, the **controller + headstage + SPI cable + electrode** go together. Add a shared computer and storage. Camera/accessories are optional. Our implant and recorder choice remain open.

Item	Price	What remains
Intan C3004 RHD controller	\$9,950 list	Includes USB/power cables; supplier confirmation needed.
Intan C3314, 32-channel headstage	\$940 list	Check electrode/implant connector.
Intan C3203 SPI cable, 0.9 m	\$215 list	One headstage connection.
Electrode/probe and adapter	Model open	Choose with the team; not in Intan subtotal.
Dell Slim ECS1250, offer ecs1250_so_16	\$549.99 offer	Windows 11, i3, 8 GB/512 GB; monitor extra.
Samsung T7 Shield 2 TB, MU-PE2T0S/AM	\$574.99 listed	Delivery unavailable on Samsung page; reference only.
Monitor / PC upgrades / backup storage	Model open	Match acquisition load and data volume.
Basler acA1300-200um camera	\$585.00	Optional body; lens, USB/I/O cables, light, and mount extra.
Camera accessories	Model open	Price after field of view and geometry are chosen.
Intan RHX software	Free	Hardware is priced separately.
Digilent WaveForms	Free	For the AD3 bench path.
RECORDER USGH	Included	Already counted with USG hardware.
Stimulus/control and analysis software	Tool open	Development time and any paid license extra.

- These are comparison prices, not selected models. Match the computer and headstage to the actual recording plan.
- Record tone markers in the neural data. Use suitable acoustic acquisition for microphone waveforms; neural inputs do not establish ultrasound capture.
- The sound, acoustic, neural, and optional video clocks still need an alignment check.

Smaller parts and subtotals

Final quantities depend on our connection diagram

For a full setup, combine **one sound group + one measurement route + one neural chain + shared computer/storage**, then add missing support parts.

Part / support	Price	How to count it
Pomona 2249-C-36 BNC cable	\$30.99 each	Example 0.9 m cable; count from wiring map.
Avisoft 52002 DIN adapter	\$48 each	USG timing connection option.
Avisoft 50021 XLR-5 cable, 2 m	\$60 each	Extra only if kit cable is unsuitable.
On-Stage DS7200B stand	\$15.95 each	Extra only if needed; clamp/adapter separate.
TDT speaker cable	Check package	May already be supplied.
Speaker wiring / output-device adapters	Model open	Match chosen amplifier and DAC connectors.
Clock/level interface, if needed	Model open	Match recorder inputs before pricing.
Fixed mounts / clamps / base / baffle	Design open	Choose after geometry; omit bundled stands.
Animal support equipment	Preparation open	Depends on later awake/sedated decision.
Enclosure / acoustic treatment	Optional	Soundproof box not required by current direction.

Partial subtotals of priced parts

Example combinations; not all to buy	Subtotal
B: NI + 40026 package + optional 40 kHz reference	\$6,162.42
NI for playback + A kit + optional 40 kHz reference	\$11,742.42
Same, using an eligible educational 116Hm kit	\$10,902.42
Intan controller + 32-channel headstage + SPI cable	\$11,105 list
Peerless + ART amplifier	\$308.46
Scan-Speak + ART amplifier	\$458.59

- Choose one measurement subtotal; do not add A and B together. Count any shared NI device once. The 40 kHz reference is optional.
- These are partial examples. Add remaining speaker, neural, computer, wiring, mounting, and band-calibration costs. Unpriced parts are not free; tax/shipping and integration are extra.

Public prices checked October 8, 2026. No supplier quote, equipment order, or measured compatibility is implied.